

१. **a. What are the requirements of ventilation? Explain the moisture movement through building components.**
b. what do you understand by thermal performance of building components? Explain the various methods of thermal insulation for exposed walls and roofs.

Answer:

a. The replacement of stale air by fresh air is known as ventilation. It is predominant contributor to thermal and hygienic environment in buildings.

The requirements of ventilation are listed below:

- To replace stale air by fresh air.
- To prevent an undue concentration of bacteria-carrying particles.
- To create healthy living conditions by preventing the undue accumulation of carbon dioxide.
- To prevent flammable concentration of gas vapor, dust and other industrial by products.
- To maintain the conditions suitable to the contents of space.

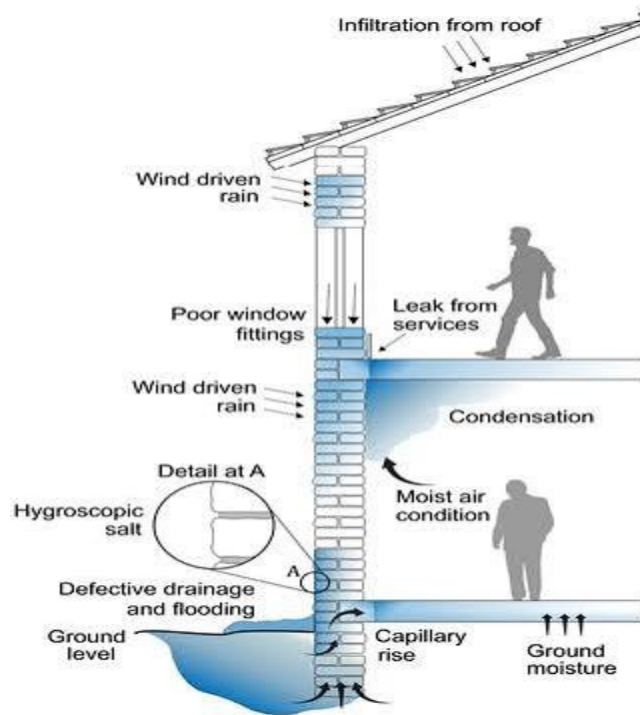


Moisture movement is the transfer of moisture in liquid or vapor form through building elements such as walls, roofs, floors, and ceilings. Excess moisture can cause dampness, mold growth, corrosion, and deterioration of building materials.

Moisture movement in buildings occur through:

- Gravity:**
 - The force of water entering by gravity is greatest on improperly sloped horizontal surfaces and vertical surfaces with penetrations.
 - These areas must remove the water from envelope surfaces through adequate sloping, correct drainage.
- Capillary action:**
 - This is the natural upward force that can draw water from source up into the cavity.
 - This occurs at the base of exterior walls.

- Building components that cannot withstand a large amount of water exposure, such as plywood, gypsum board, can create environments conducive to microbial growth or component failure.



- iii. Kinetic energy:
 - Due to wind or pressure difference forces water enters into wall openings.
 - It can be prevented by creating dam, battens etc. to interrupt the flow and redirect it to exterior.
- iv. Surface tension:
 - This allows water to adhere to and travel along the underside of building components such as joints, and window heads.
 - This water can be drawn into the building by gravity or unequal pressures.
- v. Vapor diffusion:
 - Water vapor moves from areas of high vapor pressure (high humidity) to low vapor pressure through porous materials.
 - Controlled by using vapor barriers.
- vi. Air leakage:
 - Moist air enters or leaves through cracks, joints, doors, windows, and gaps.
 - One of the major causes of condensation inside buildings.

b. The ability of building components such as walls, roofs, floors, windows, etc. to control the transfer of heat between the indoor and outdoor environments is known as thermal performance of buildings.

Good thermal performance keeps indoor comfortable for staying by reducing heat gain in summer and heat loss in winter, while minimizing energy consumption.

The various method of thermal insulated for exposed wall and roof are stated below:

i. Exposed wall:

- a. Construction of cavity wall:
 - Two walls are constructed separated by an air gap or cavity in between,
 - Air cavity acts an insulating layer and reduces heat transfer.

- b. Insulating boards:

- Insulating boards of the materials such as polystyrene, extruded polystyrene or mineral wool is fixed to walls.
- It provides excellent thermal resistance.
- c. Reflective surface coating:
 - Light-colored or reflective paints can be used to reduce the solar heat absorption.
 - The heat transfer is reduced due to low thermal conductivity.
- d. Lightweight insulating blocks:
 - Lightweight insulating blocks such as Autoclaved Aerated Concrete blocks or hollow concrete blocks can be used.
 - It provides excellent thermal resistance.

ii. Exposed roof:

- a. False ceiling:
 - A ceiling made up of gypsum board, PVC or fiberboard is provided as false ceiling.
 - It creates an insulating air space below the roof.
- b. Roof terrace insulation:
 - Insulating materials are placed above the roof slab and is protected by screed or tiles.
- c. Reflective roof:
 - White or reflective coatings can be provided on the roof reduces the absorption of solar radiation.
- d. Ventilated roof:
 - Air space between the roof covering and ceiling allows hot air to escape, reducing the indoor temperature.

2. a. Define foundation. Describe the types of foundation with necessary sketches.

b. What are the types of flooring? Explain the process of terrazzo finish floor.

Answer:

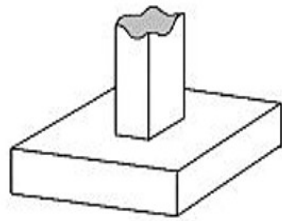
a. Foundation is the lowermost part of structure which receives load from superstructure and transfers it to stable soil layer. The principal loads are:

1. **Dead Load:**
2. **Live (Imposed) Load:**
3. **Wind Load:**
4. **Earthquake (Seismic) Load:**
5. **Impact and Vibratory Loads:**

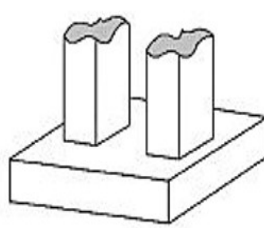
Foundations are broadly classified into two categories:

a. Shallow foundations:

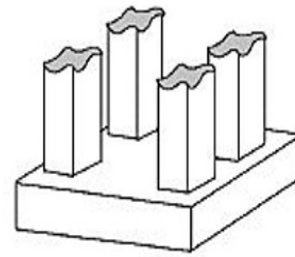
- Used when the soil with adequate bearing capacity is available near the ground surface (generally depth $\leq$ width).



Isolated Footing



Continuous footing



Mat footing

Types:

a. Strip (Wall) Footing:

- The footing that runs along the wall i.e., wall footing or continuous footing is known as strip footing.
- It is the slab supporting wall.

b. Isolated (Pad) Footing:

- Isolated footing is the slab supporting columns.
- It is constructed in RCC framed residential buildings.

c. Combined Footing:

- It is the slab supporting two or more columns in a line.
- Combined footing is provided to support columns at property line and closely spaced columns.

d. Strap (Cantilever) Footing:

- Strap footing is provided with a beam at foundation level to connect two columns which is known as strap beam.
- It is provided to support the column at property line.

e. Raft (Mat) Foundation:

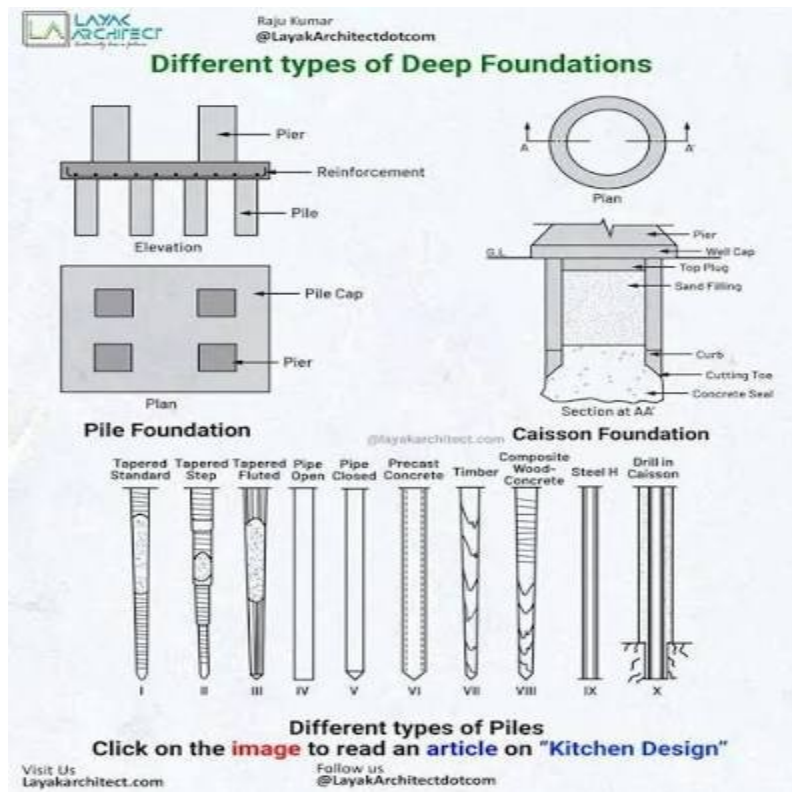
It is the slab supporting large number of columns in two or more rows.

It is used in compressible soil layer where isolated footings don't perform well and commercial buildings where isolated footing covers more than 50% of the plinth area.

Applications: Residential buildings, low-rise structures, and light to moderate loads.

b. Deep foundations:

Used when the upper soil is weak and stronger strata are available at greater depths (depth > width).



Types:

a. Pile Foundation:

- Pile foundation consists of long slender member called pile that transfer the load to the ground through bearing and skin friction.
- Piles also resist the structures against uplift and provide structural stability against lateral and overturning forces.

b. Pier Foundation:

- Pier foundations are those foundations which transfer loads that cannot be carried by shallow foundations.
- The foundation transfers the load to the ground only by bearing.
- When the top most soil is stiff clay, it resists the driving of bearing pile and hence in that case, pier foundation is used.

c. Caisson (Well) Foundation

- Well or caisson foundation is the watertight structure used in bridge pier, dam construction etc.
- It is a readymade hollow cylinder depressed into the soil up to the desired level and then filled with concrete that converts into foundations.

Applications: High-rise buildings, bridges, waterfront structures, and heavy industrial facilities.

b. Floors are the horizontal elements of a building structure which divide the building into different level for the purpose of creating more accommodation within a restricted space one above the other and provide support to occupants.

The types of flooring are described below:

i. Mud and muram flooring:

- Used only in low-cost housing in rural areas.

- ii. Brick flooring:
 - Suitable for cheap construction such as ware houses, stores where good bricks are available.
- iii. Cement concrete flooring:
 - Commonly used for residential, commercial, industrial buildings which is moderately cheap, durable and easy to construct.
- iv. Terrazzo flooring:
 - Has very decorative and good wearing properties which is expensive and is best suited for outdoor spaces such as verandas.
- v. Mosaic flooring:
 - Made up of broken pieces of tiles of china glazed or cement, or marble arranged in different patterns which becomes slippery when polished.
- vi. Tiles flooring:
 - Tile flooring are durable, moisture resistant and easy to clean, which is widely used in bathrooms, kitchen and entry ways.
- vii. Marble flooring:
 - It is a superior type of flooring used in bathroom, kitchen and comes in many colors and sizes.
- viii. Timber flooring:
 - It is used for dancing halls, auditorium, sitting rooms etc. but has to be protected from dampness.
- ix. Asphalt flooring:
 - This type of flooring is dustless, elastic, water proof, acid proof which is made up of sand or grit mixed to melted asphalts in proportion of 2:1.
- x. Glass flooring:
 - It is a special type of flooring used where it is desired to transmit light from upper floor to lower floor.

Terrazzo floor finish:

- This is a composite material made up of cement and marble chips aggregate 3 mm to 6 mm size in proportion 1:1.25 to 1:2.
- It is then mixed and poured in-situ onto a concrete base.
- The chips are exposed by grinding. The surface is then cleaned and wax polish is applied to get final shiny surface.
- It can also be premade and is available in slabs or tiles.
- It is very hard wearing, if it is polished or if wet is very slippery.
- It is best for outdoor spaces such as verandas.



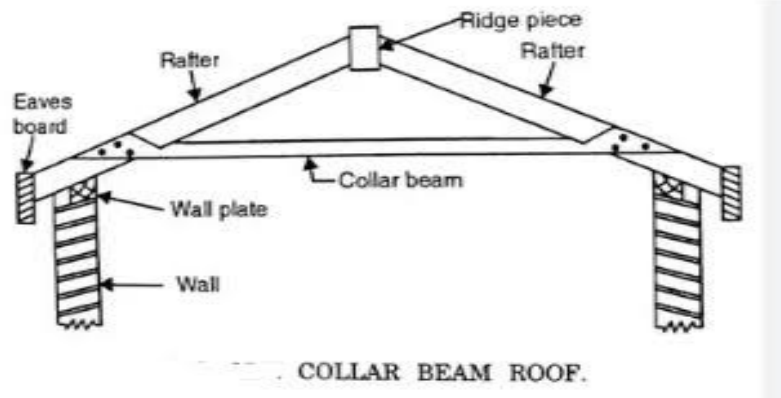
- Procedure:
 - i. Preparation of base:
 - The base is cleaned thoroughly.
 - The surface is leveled, cracks are repaired and roughen to improve bonding.
 - ii. Laying the underlayer:
 - Cement concrete or cement mortar base, 20-40 mm thick is laid, compacted and leveled properly.
 - Brass, aluminum or glass strips are installed according to design.
 - iii. Preparation of terrazzo:
 - Ordinary Portland cement is mixed with marble chips and other aggregates with pigments if color is to be added as well as water to obtain workable mix.
 - iv. Laying of toppings:
 - Terrazzo mix is spread in about 10-15 mm thickness between the divider strips.
 - The mix is compacted using roller or trowel.
 - Marble chips are then uniformly distributed.
 - v. Curing:
 - The floor is cured for 5-7 days by sprinkling water.
 - Strength is developed after proper curing and prevents shrinkage cracks.
 - vi. Grinding:
 - The first grinding is performed using a coarse abrasive stone after terrazzo has hardened.
 - Pinholes or voids are filled with matching cement slurry.
 - Second and final grinding is carried out with finer abrasives until the surface becomes smooth.
 - vii. Polishing:
 - The floor is polished using polishing powder or polishing machines.
 - Wax or a sealing compound must be applied to enhance shine and surface protection.
 - viii. Final cleaning:
 - Terrazzo floor is thoroughly washed and cleaned after which it is ready to use.

3. **a. Draw and explain different components of timber collar beam roof with their sizes. Explain the different types of roofs covering for pitched roof.**
b. Explain the preparation of cement sand mortar (1:6). Differentiate between random rubble, coursed rubble and ashlar stone masonry with figure.

Answer:

a.

- A timber collar beam roof is a pitched roof in which a horizontal collar beam connects a pair of rafters near the upper part of the roof.
- It prevents the rafters from spreading and increases the roof's stability.
- It is commonly used for spans of 4.5–8 m.



Components of timber collar beam roof:

- i. Ridge piece:
 - It supports the upper ends of rafters at the roof apex.
 - The size of ridge piece is about 100 x 25 to 150 x 38 mm.
- ii. Rafters:
 - It is the main inclined member carrying the roof covering loads to walls.
 - The size of rafters range about 100 x 75 to 150 x 75 mm.
- iii. Collar beam:
 - It connects the opposite rafters and prevents from sagging and lateral movement.
 - The size of collar beam ranges from 100 x 75 to 150 x 100 mm.
- iv. Wall plate:
 - It distributes the roof load uniformly to supporting wall.
 - The size of wall plate ranges from 100 x 75 to 150 x 100 mm.
- v. Purlins:
 - Purlins are the horizontal members supporting the rafters or roof covering.
 - The size of purlin ranges from 100 x 75 to 150 x 100 mm.
- vi. Batten:
 - They support the roofing materials such as tiles or slates.
 - The size of batten ranges from 50 x 25 to 75 x 25 mm.
- vii. Struts:
 - They support the rafters and reduce bending in long span roofs.
 - The size of struts ranges from 75 x 75 to 100 x 100 mm.
- viii. Braces:
 - Braces provides the lateral stability against wind loads.
 - The size ranges from 75 x 50 mm.
- ix. Roof covering:
 - It protects the building from weather such as CGI sheets, tiles, slates etc.
 - The thickness of roofing materials depends upon the materials chosen.

The different types of roof covering are listed below:

- i. Thatch:
 - Thatch is dry vegetation made up of straw, water reed, sedge or heather.
 - It is simply tied tightly in a bunch of thickness 10 to 15 cm on the battens in closely distanced about 30 cm in layers.

- This type of roof is widely used in rural areas.



ii. Tile roofing:

- Tile roofing are made up of ceramic material and is hard and brittle.
- They are manufactured from clay and concrete to a wide range of design and colors suitable for the pitches from 20° to 45° .



iii. C.G.I sheet roofing:

- The rafter is fixed to wall plate and ridge and purlin laid on it.
- C.G.I sheets are nailed or bolted to purlins.
- Eaves board is nailed to the ends of rafters.



iv. Asbestos Cement sheet roofing:

- AC sheet are known as fiber cement sheet.
- They are produced by the mixture of cement, asbestos fibers and water,
- They are cheap, light, durable and fire resistant.



v. Slate roofing:

- Slates are fine-grained, foliated metamorphic rock derived from shale type of sedimentary rock.
- They are nailed to battens that are fixed on the planks or directly on rafters.
- They are heavy and requires a strong roof truss.



b.

A cement sand mortar of 1:6 consists of 1 part of cement and 6 parts of sand by volume.

The preparation of cement sand mortar of 1:6 is explained below:

Materials required:

- i. Ordinary Portland Cement
- ii. Well graded sand
- iii. Clean potable water
- iv. Mixing platform

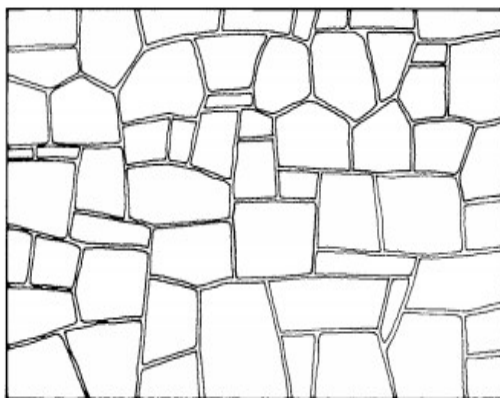
v. Measuring boxes

Procedures:

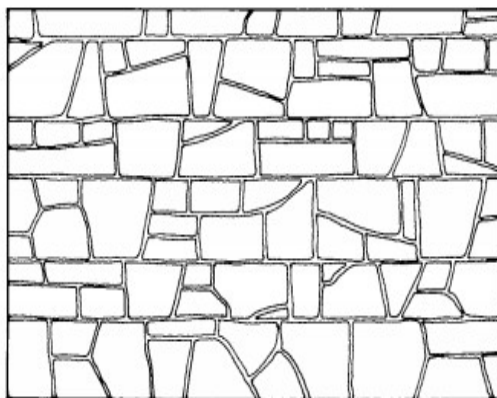
- i. The ingredients are measured in which 1 part is cement and 6 part is sand, accurately proportioned using measuring boxes.
- ii. The mixture is spread in the mixing platform and is dry mixed until the color becomes uniform.
- iii. The mixture is then formed into a heap and a small depression is made at the center.
- iv. Clean water is added gradually while mixing.
- v. The process of mixing is continued until a uniform, plastic, workable mortar is obtained.
- vi. There must not be excess water or no lumps of cement or dry patches in the mixture.
- vii. The mortar must be used within 30 minutes after addition of water, before the initial setting period of cement.

The differences between random rubble, coursed rubble and ashlar stone masonry are listed below:

Random rubble masonry	Coursed Rubble masonry	Ashlar masonry
Random rubble masonry consists of irregular, undressed or roughly dressed stones.	Coursed rubble masonry consists of roughly dressed stones of nearly equal height.	Ashlar masonry consists of finely dressed rectangular block stones.
This type of masonry is thick and irregular with thickness of 10-20 mm or more.	This type of masonry is of moderate thickness of 10-15 mm.	This type of masonry is very thin of thickness 3-6 mm.
They look rough and irregular.	They look fairly neat.	They look very neat and attractive.
Construction cost of random rubble masonry is lowest.	Construction cost of coursed rubble masonry is moderate.	Construction cost of Ashlar masonry is highest.
It requires less skilled labor for construction.	It requires semi-skilled labors for construction.	It requires highly skilled labors for construction.
Time required for construction is fast.	Time required for construction is moderate.	Time required for construction is slow.
This type of masonry is used in retaining walls, foundations, boundary walls.	This type of masonry is used in residential buildings, compound walls etc.	This type of masonry is used in monuments, bridges, public and prestigious building.



Random Rubble



Coursed Rubble

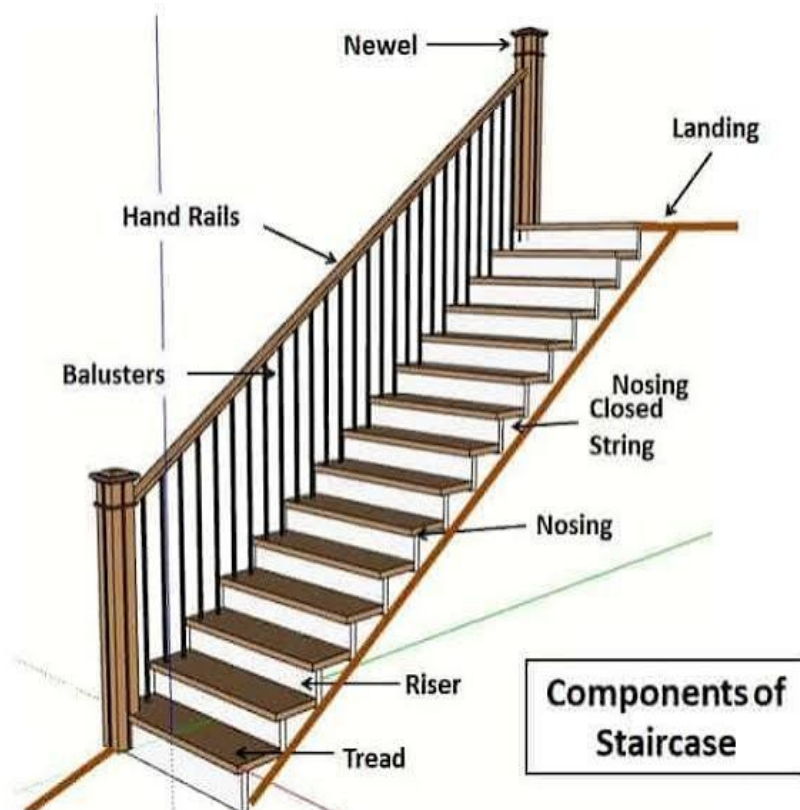


4. a. Define stair. Illustrate the elements of staircase with figure.
 b. With the help of neat sketches, differentiate between solid and suspended ground floor.

Answer:

a.

- A stair is a convenient means of access between the floors of building constructed to provide easy, comfortable and safe ascent and descent.
- It consists of series of steps that are neither laborious nor difficult to climb within an enclosure called staircase.
- The major functions of stair are:
 - i. To provide normal everyday access from floor to floor.
 - ii. To provide an easy and rapid escape, from upper floor in case of emergency such as fire, earthquakes etc.
- Elements of staircase:

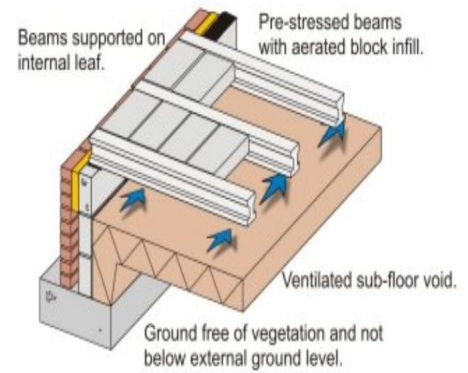
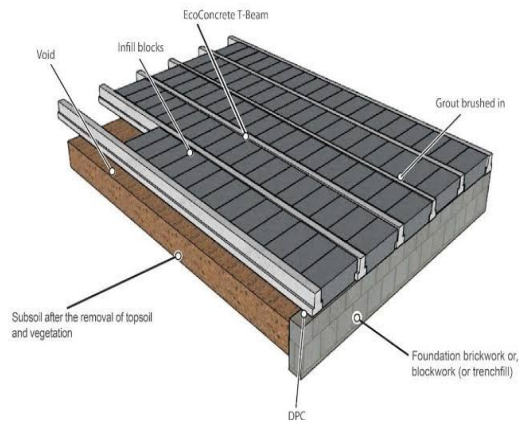


- i. Tread:
 - It is the upper surface of a step on which foot is placed during ascents and descents.
- ii. Riser:
 - It is the vertical member between two successful treads.
- iii. Nosing:
 - It is the exposed edge of a tread, usually projecting with a square, rounded or splayed edge.
- iv. Going:
 - It is the distance measured horizontally from face of one riser to the face of next riser.
- v. Flight:
 - Flight is an uninterrupted series of steps between floors or between floor and landings or between landing and landing.
- vi. Landing:
 - It is a flat platform or resting place provided at the head of a series of steps.
- vii. Steps:
 - A single unit of a stair consisting of one tread and one riser can be called step.
- viii. Handrails:
 - It is the member of stair at the top of baluster and act as support of the hand.
- ix. Newel:
 - A wooden, steel post supporting hand rail and is usually provided at the starting of flight.
- x. Baluster:
 - Vertical member to support the handrail and placed between steps and handrails.

b. The differences between solid ground and suspended ground floor are listed below:

Solid ground	Suspended ground floor
It is constructed directly on well-compacted ground.	It is constructed above the ground supported by beams, walls or joists.
Floor load is transferred directly to the ground.	Floor load is transferred to supporting walls, beams or columns.
It is suitable where the soil has good bearing capacity and is dry.	It is suitable where the ground is weak, damp, expansive or flood prone.
They are less expensive to construct.	They are more expensive due to additional structural members.
Simple and quick construction.	More complex construction.

Typical Ground Floor Slab

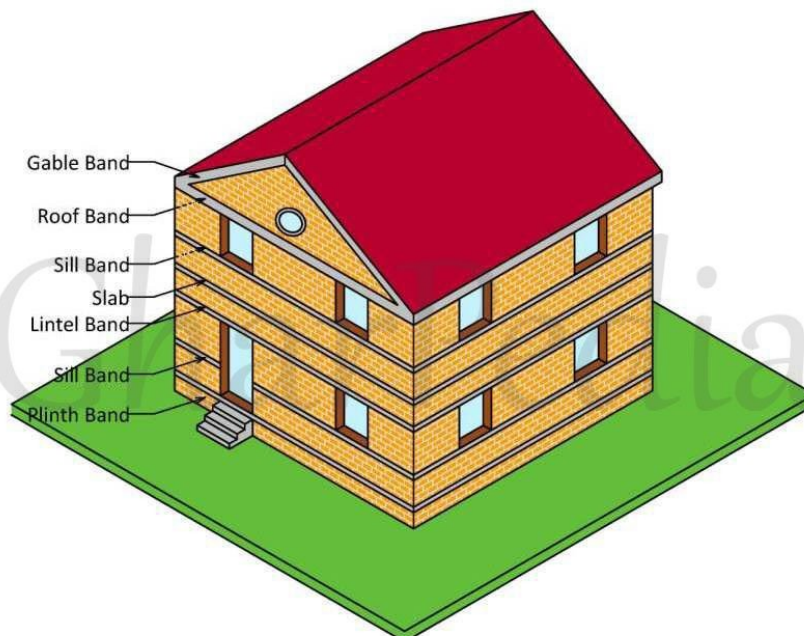


5. **a. How do you make a brick masonry buildings earthquake resistant? Explain with sketches various measures adopted.**
b. Illustrate components of the rooftop rain water harvesting system.

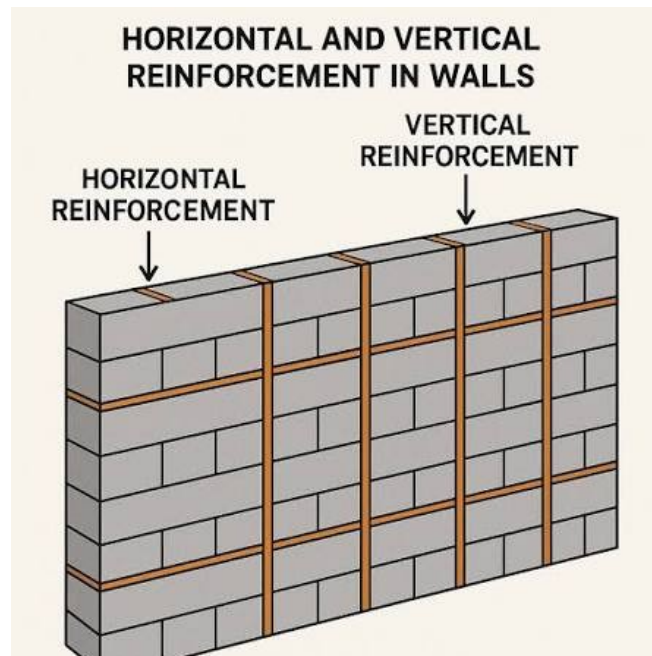
Answer:

a.

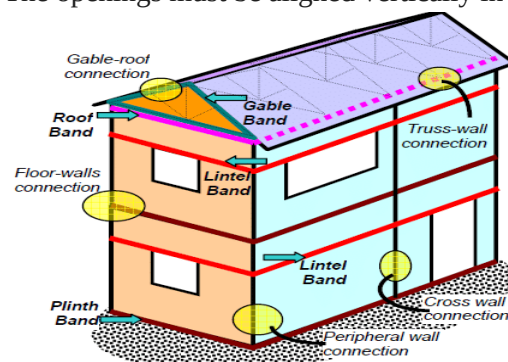
- Brick masonry buildings are weak in tension and are brittle in nature which makes them vulnerable during earthquakes.
- To improve their seismic performance, special earthquake-resistant construction measures are adopted to increase strength, ductility, and stability.
- Brick masonry buildings can be made earthquake resilient by:
 - i. Providing horizontal seismic bands:
 - Horizontal reinforced concrete bands tie all the walls together and prevent separation during earthquakes.
 - Different bands such as plinth band, sill band, gable band, lintel band must be provided to make the brick masonry earthquake resistant.



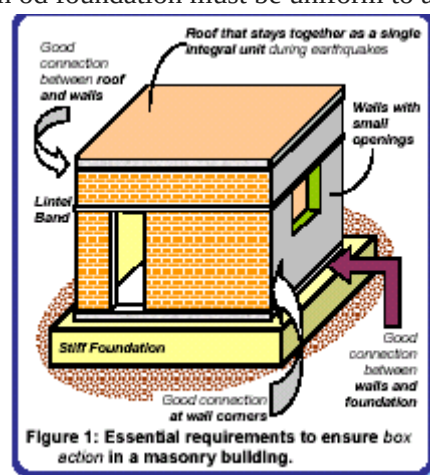
- ii. Providing vertical reinforcement:
 - Steel bars can be incorporated in the brick masonry at building corners, wall junctions and both side of window and door openings.



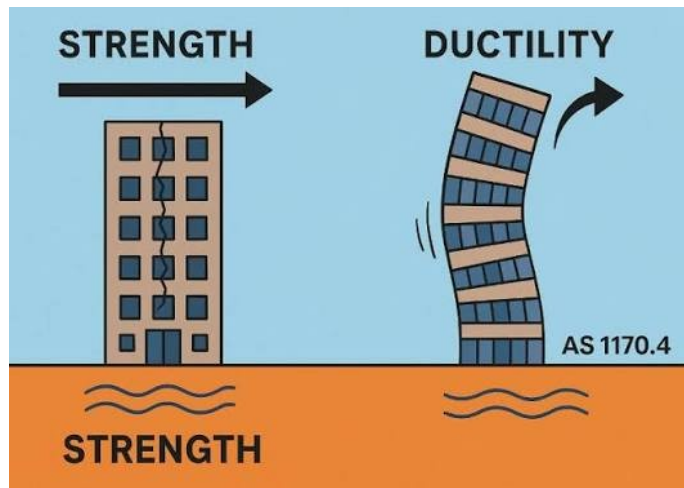
- iii. Size and location of openings:
 - Doors and window openings should not be too large or too small.
 - The openings must be aligned vertically in symmetry in different floors.



- iv. Strong foundation:
 - The foundation of building must be constructed on firm soil.
 - The depth of foundation must be uniform to avoid differential settlement.

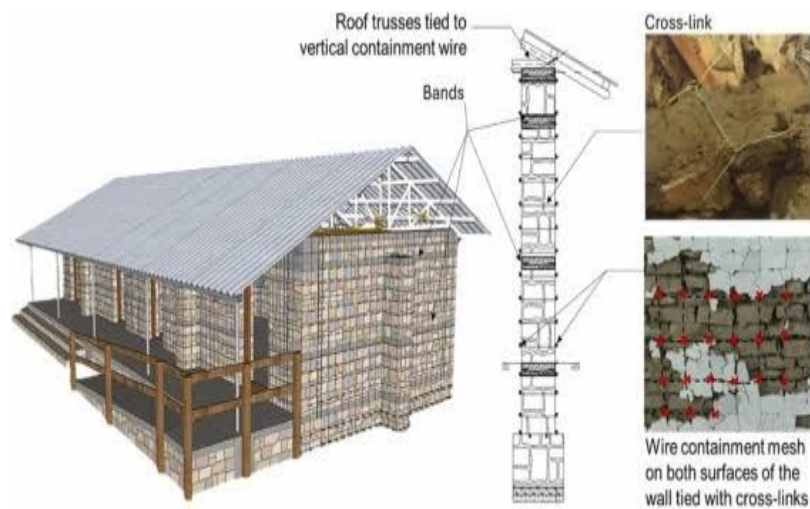


- v. Improving ductility:
 - Buildings can be made ductile by proper detailing.

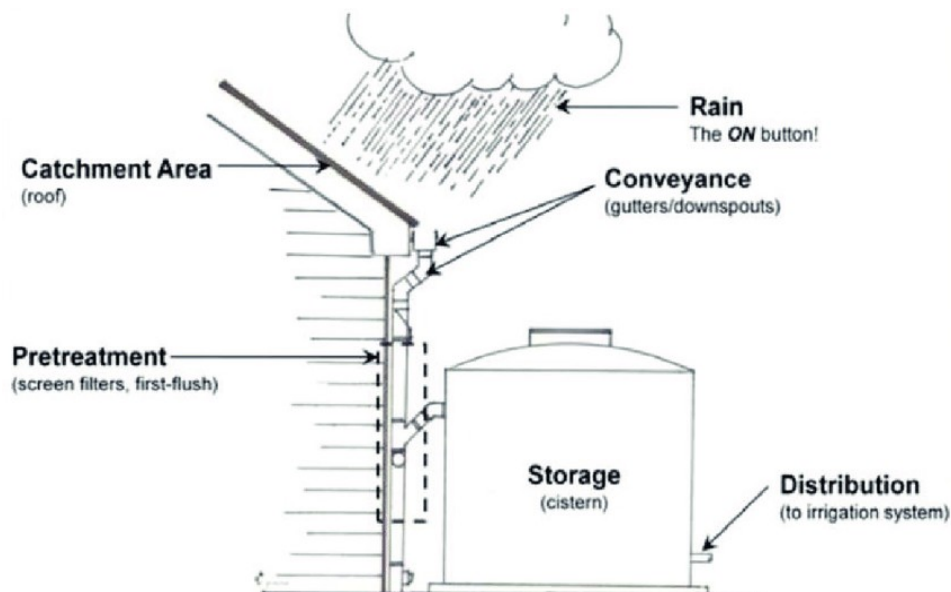


vi. Light roof construction:

- Lightweight roofing materials such as C.G.I sheet are used.
- The roof must be properly anchored to the wall.



b. The components of rooftop rainwater harvesting system are described below:



i. Catchment area:

- Catchment area for rooftop rainwater harvesting system is basically the roof, that receives rainfall.
- The quantity of water collected depends upon the rainfall and roof area.
- ii. Gutters:
 - Gutter channels are fixed along the edge of roof.
 - They collect rainwater from the roof and convey it to downpipes.
- iii. Downpipe:
 - Downpipe is vertical pipe that carries water from gutter channel to the filter or storage tank.
 - PVC pipes are commonly used as a downpipe.
- iv. First flush device:
 - This device diverts the initial rainwater containing dust, leaves and other impurities.
 - It improves the quality of harvested water.
- v. Filter unit:
 - Filter unit filters the water and removes suspended impurities before storage.
 - It consists of gravel, sand, charcoal, pebbles and filter mesh.
- vi. Storage tank:
 - Storage tank finally stores the filtered rainwater.
 - It can be made up of plastic, RCC or masonry.
- vii. Distribution system:
 - This system consists of taps, pumps, pipelines used to distribute stored water for domestic or irrigation purposes.
- viii. Overflow pipe:
 - They carry excess water safely away when the storage tank is full.
- ix. Recharge well:
 - It allows excess rainwater to infiltrate into the ground and replenish groundwater.